

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT NAME: PERFORM PROGRAMMING PRINCIPLES
UNIT CODE: 0613 451 05A

CONTINUOUS ASSESSMENT TEST (CAT) –CAT 2

Time allowed: 1 hour 30 minutes Total marks: 50

Name: _____

Admission no.: _____

Class: _____

Date: _____

Instructions to the candidate

- ✓ Answer ALL questions.
- ✓ The marks for each question are shown in brackets.
- ✓ Write all programs neatly in C and include comments where possible.
- ✓ No mobile phones or notes are allowed in the examination room.

QUESTIONS

1. Define the term control structure and name the three main types of control structures in C. (3 marks)

2. Name three input functions in C and state what each one reads. (3 marks)

3. State four criteria used when selecting a data type based on system requirements. (4 marks)

4. Differentiate between a function declaration (prototype) and a function definition. (3 marks)

5. Study the array declaration below and answer the questions that follow. (3 marks)

```
int marks[5] = {50, 60, 70, 80, 90};
```

(a) State the value stored in marks[2] and marks[4]. (2 marks)

(b) State the first and last valid index of the array. (1 mark)

6. A three-dimensional array is declared as shown below. (4 marks)

```
int a[2][3][4];
```

(a) Explain the meaning of each of the three indexes. (3 marks)

(b) Calculate the total number of elements the array can store.

(1 mark)

7. Study the two-dimensional array below. (5 marks)

```
int scores[3][4] = {  
    {12, 8, 15, 20},  
    {7, 18, 9, 11},  
    {14, 6, 19, 5}  
};
```

(a) State the number of rows and the number of columns in the

array. (2 marks)

(b) Write the values stored in scores[1][1] and scores[2][3].

(2 marks)

(c) Write a C statement that changes the element in row 2,
column 0 to 30. (1 mark)

8. Study the two programs below. (5 marks)

Program 1

```
#include <stdio.h>
```

```
void change(int x)
```

```
{
```

```
    x = 50;
```

```
}
```

```
int main()
```

```
{
```

```
    int number = 10;
```

```
    change(number);
```

```
    printf("Number = %d", number);
```

```
    return 0;
```

```
}
```

Program 2

```
#include <stdio.h>
```

```
void change(int *x)
{
    *x = 50;
}
```

```
int main()
{
    int number = 10;
    change(&number);
    printf("Number = %d", number);
    return 0;
}
```

- (a) State the output of Program 1 and of Program 2. (2 marks)
- (b) Name the method of passing arguments used in each program.
(2 marks)
- (c) Explain why the outputs of the two programs are different.
(1 mark)

9. Select the most suitable C data type for storing each of the following data, giving one reason to justify each choice. (5 marks)

- (a) A student's age
- (b) The population of a country (more than 2.1 billion)
- (c) A percentage score from 0 to 100 on a device with very limited memory
- (d) A measurement requiring about 15 digits of precision in a scientific calculation
- (e) A single grade letter such as 'A'

10. Write a complete C program that accepts a student's marks from the keyboard and uses an if...else if structure to display the grade as follows: (7 marks)

80 and above: Grade A

60 to 79: Grade B

50 to 59: Grade C

Below 50: Fail

11. Write a complete C program that: (8 marks)

(a) Defines a user-defined function called multiply that accepts two integers and returns their product. (3 marks)

(b) Includes a function prototype for the function. (1 mark)

(c) In the main function, accepts two integers from the user using scanf(), calls the function, and displays the result. (4 marks)